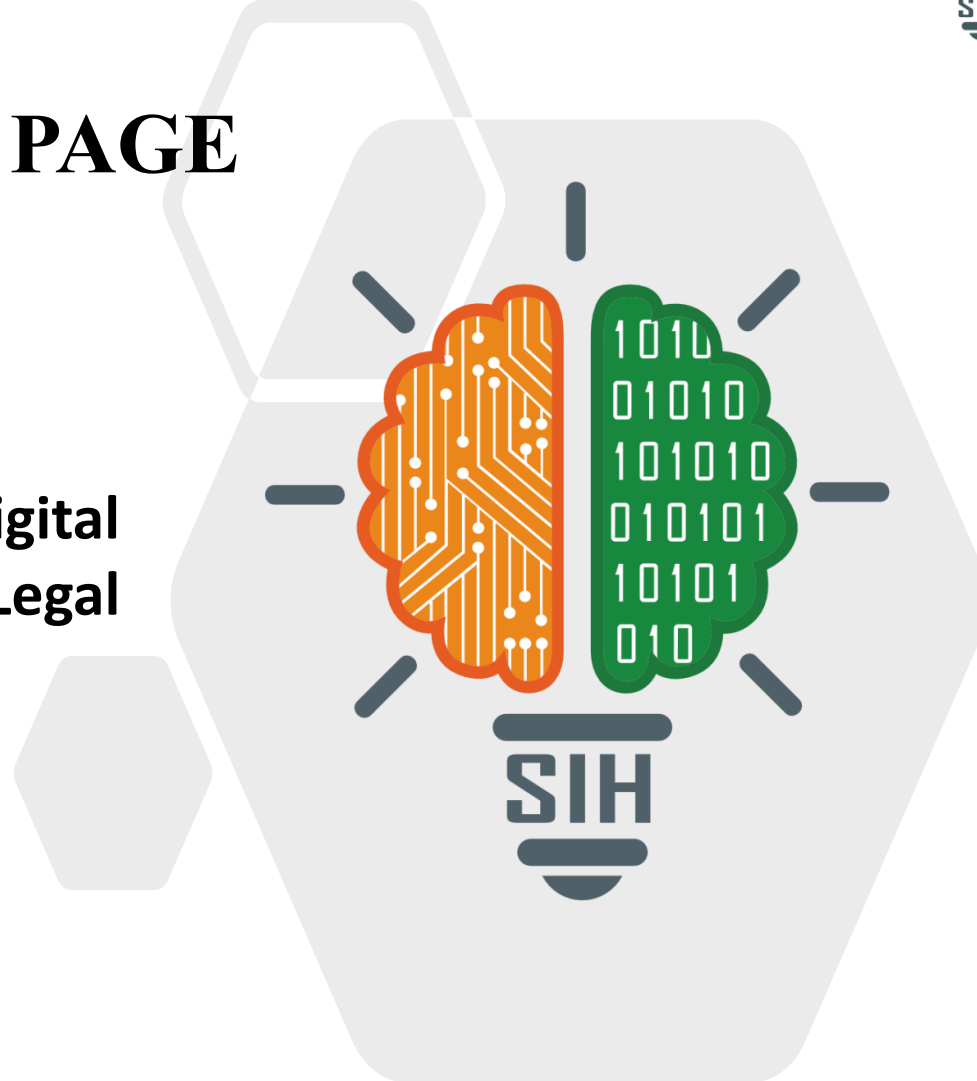


TITLE PAGE

- **Problem Statement ID – SIH26190**
- **Problem Statement Title- Secure Digital Document Management System for Legal and Investigation Documents**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID- S0055**
- **Team Name- CipherShield**



IDEA TITLE

**PROBLEMS FOUND**

1



Unauthorized Access
Sensitive documents may be accessed by users without proper authorization.

→



Solution

- JWT authentication
- Bcrypt password hashing
- Employee ID support

2



Data Leakage
Confidential information can be copied, downloaded or shared improperly.

→



Solution

- Role-based access control (RBAC)
- Restricted permissions
- Tamper-proof access logs

3



Document Can't Modify After Upload
Documents may be modified after uploading without authorization.

→



Solution

- AES-256 encryption
- SHA-256 hash generation
- Immutable storage + audit trail

6



Poor Traceability
Difficult to determine who accessed what, when and what action was performed.

→



Solution

- Complete audit logs
- Event-based tracking
- Hash-chained activity records

**END-TO-END PROCESS FLOW**

1



User Login & Authentication
(JWT + Bcrypt + Employee ID)

↓

2



Role-Based Access Control (RBAC) Check
(Admin / Officer / Legal Reviewer / Clerk / Auditor)

↓

3



Create / Select Investigation Case
(CASE-2026-XXXXX + Confidentiality Tagging)

↓

4



Evidence & Document Upload
(Multipart FormData + Multer)

↓

5



AES-256 Encryption + SHA-256 Hash
(Digital fingerprint generation)

↓

6



Immutable Secure Storage
(Appwrite + MongoDB)

↓

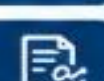
7



Multi-Stage Legal Review & Approval
(Judicial / Legal Review Portal)

↓

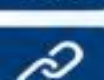
8



Digital Signature & Access Control
(Watermarking + Restricted Access)

↓

9



Hash-Chained Audit & Activity Tracking
(EventHash + PreviousHash)

↓

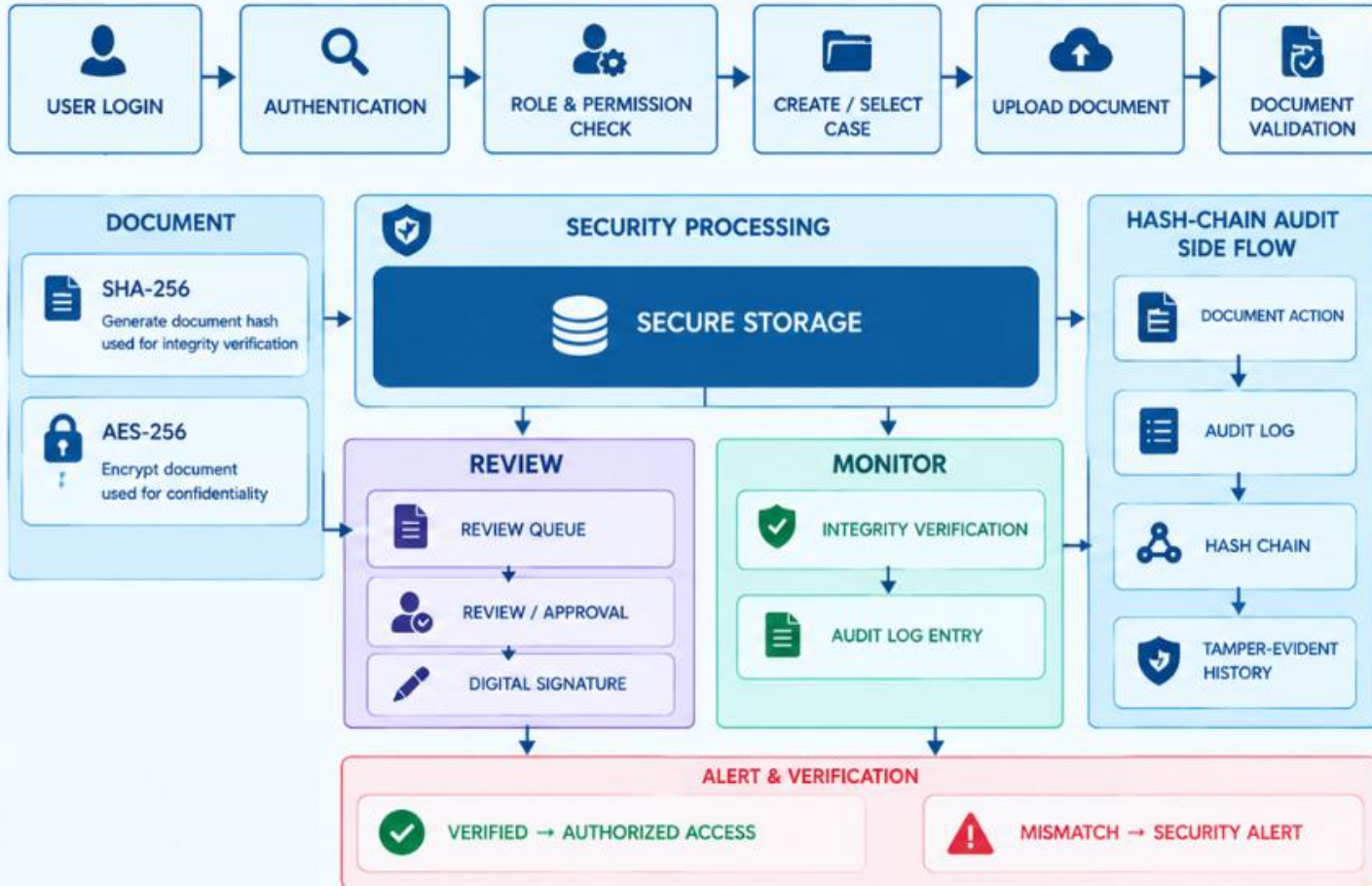
10



Real-Time Integrity Verification
(Stored Hash = Calculated Hash)

1 PROCESS FLOW DIAGRAM

Complete document lifecycle in SecureDocs — access, security, review & monitoring



2 TECHNOLOGY STACK

1 - FRONTEND

React 19
TypeScript
Vite 7
Tailwind CSS 4

2 - BACKEND

Node.js 22
Express.js 5
TypeScript

3 - DATABASE

MongoDB 7.0
Mongoose 9

4 - UI & DATA TOOLS

shadcn/ui
Lucide Icons
Recharts
React Query
React Hook Form
Zod

5 - AUTHENTICATION & AUTHORIZATION

JWT Authentication
bcryptjs
Role-Based Access Control (RBAC)

6 - FILE STORAGE

Firebase Storage
Local storage fallback

7 - DEPLOYMENT

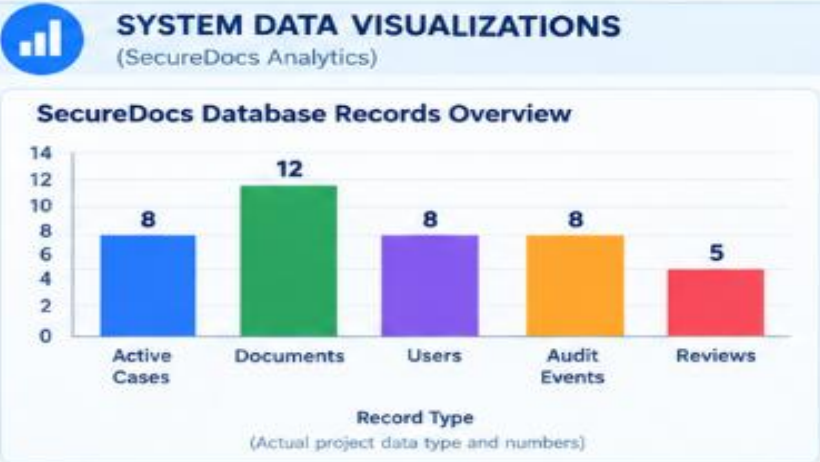
Vercel

- Frontend: Static Files (Vercel CDN)
- Backend: Serverless Functions (/api/*)
- Database: MongoDB Atlas

Docker + Nginx (Alternative)
Self-hosted deployment (not used for live production)

8 - SECURITY

SHA-256 Document Integrity
Hash-Chained Audit Logs





LAW ENFORCEMENT AGENCIES

- Secure and organized access to investigation documents
- Faster case processing and coordination
- Reduced risk of document tampering
- Better tracking of document lifecycle



LEGAL DEPARTMENTS & COURTS

- Quick and reliable access to case files
- Ensures document authenticity and validity
- Simplifies review and verification process
- Supports faster and fairer justice



AUDIT & COMPLIANCE TEAMS

- Complete audit trail and activity logs
- Detects unusual access patterns
- Ensures accountability and transparency
- Helps in regulatory compliance



GOVERNMENT & ORGANIZATIONS

- Improved operational efficiency
- Reduced manual effort and errors
- Stronger data security and integrity
- Builds public trust and institutional credibility

FROM SECURE DOCUMENTS TO A STRONGER JUSTICE SYSTEM



**DIGITIZE &
SECURE DOCUMENTS**



**ENCRYPT &
PROTECT DATA**



**ENABLE ROLE-BASED
ACCESS**



**TRACK & LOG
ALL ACTIVITIES**



**ENSURE INTEGRITY
& AUTHENTICITY**



**SUPPORT FASTER,
FAIRER JUSTICE**



ENHANCED SECURITY

Protects sensitive information with encryption and tamper-proof mechanisms.



HIGHER EFFICIENCY

Reduces manual work and speeds up document processing.



IMPROVED TRANSPARENCY

Full audit trail and real-time monitoring build trust and accountability.



BETTER COLLABORATION

Enables seamless coordination across agencies, departments and stakeholders.



LONG-TERM IMPACT

Strengthens the justice system and supports a safer, more transparent society.

Standards, frameworks and documentation underpinning the design

	NIST SP 800-53 Standards / Control	Security and audit control guidelines
	OWASP Security best practice	Secure file handling and access control best practices
	FIPS 180-4 (SHA-256) Cryptography	Cryptographic hashing standard used for integrity verification
	Firebase / Google Cloud Platform documentation	Authentication, Firestore and Storage security rules

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.